

ECE/CS 757 Advanced Computer Architecture II

Spring 2025

Consistency (Part 4)

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Some slides adopted from Greg Steffan, Natalie Enright Jerger and Mikko Lipasti

Last Lecture

- Relaxed consistency
- Data-race-free
- DeNovo coherence

This Lecture

- Relaxed consistency examples
- Interconnects
- Network diameter
- Channel load and bisection bandwidth
- Path diversity

Relaxed Consistency

S. V. Adve and K. Gharachorloo, *Shared Memory Consistency Models: A Tutorial*, Western Research Laboratory Technical Report, 1995.

Relaxation	$W \rightarrow R$ Order	$W \rightarrow W$ Order	$R \rightarrow RW$ Order	Read Others' Write Early	Read Own Write Early
SC [16]					✓
IBM 370 [14]	✓				
TSO [20]	✓				✓
PC [13, 12]	✓			✓	✓
PSO [20]	✓	✓			✓
WO [5]	✓	✓	✓		✓
RCsc [13, 12]	✓	✓	✓		✓
RCpc [13, 12]	✓	✓	✓	✓	✓
Alpha [19]	✓	✓	✓		✓
RMO [21]	✓	✓	✓		✓
PowerPC [17, 4]	✓	✓	✓	✓	✓

Relaxed Consistency

Possible with SC: A=0, B=0, C=0, D=0, E=0, F=0?

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

i.

A	B	C	D	E	F	SC?
0	0	0	0	0	0	

Relaxed Consistency

Possible with SC: A=0, B=0, C=0, D=0, E=0, F=0?

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;  
A = *y;
```

```
// P2
```

```
*y = 1;  
B = *x;
```

```
// P3
```

```
C = *x;  
D = *y;
```

```
// P4
```

```
E = *y;  
F = *x;
```

i.

A	B	C	D	E	F	SC?
0	0	0	0	0	0	

Relaxed Consistency

Possible with SC: A=0, B=0, C=0, D=0, E=0, F=0? No

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

i.

A	B	C	D	E	F	SC?
0	0	0	0	0	0	no

Relaxed Consistency

Possible with TSO: A=0, B=0, C=0, D=0, E=0, F=0?

- Relaxed write-to-read, with store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

i.

A	B	C	D	E	F	SC?	TSO?
0	0	0	0	0	0	no	

Relaxed Consistency

Possible with TSO: A=0, B=0, C=0, D=0, E=0, F=0?

- Relaxed write-to-read, with store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

relaxed write to read

// P1 // P2



```
*x = 1;                      *y = 1;
```

```
A = *y;                      B = *x;
```

// P3

```
C = *x;
```

```
D = *y;
```

// P4

```
E = *y;
```

```
F = *x;
```

i.

A	B	C	D	E	F	SC?	TSO?
0	0	0	0	0	0	no	

Relaxed Consistency

Possible with TSO: A=0, B=0, C=0, D=0, E=0, F=0? **Yes**

- Relaxed write-to-read, with store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

relaxed write to read

// P1 // P2



```
*x = 1;                      *y = 1;
A = *y;                      B = *x;
```

// P3 happens first // P4

```
C = *x;                      E = *y;
D = *y;                      F = *x;
```

i.

A	B	C	D	E	F	SC?	TSO?
0	0	0	0	0	0	no	yes

Relaxed Consistency

Possible with PC: A=0, B=0, C=0, D=0, E=0, F=0?

- Relaxed write-to-read, without store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

i.

A	B	C	D	E	F	SC?	TSO?	PC?
0	0	0	0	0	0	no	yes	

Relaxed Consistency

Possible with PC: A=0, B=0, C=0, D=0, E=0, F=0? **Yes**

- Relaxed write-to-read, without store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

relaxed write to read

// P1 // P2



```
*x = 1;                      *y = 1;
```

```
A = *y;                      B = *x;
```

// P3 happens first // P4

```
C = *x;                      E = *y;
```

```
D = *y;                      F = *x;
```

i.

A	B	C	D	E	F	SC?	TSO?	PC?
0	0	0	0	0	0	no	yes	yes

Relaxed Consistency

Possible with SC: A=0, B=0, C=1, D=0, E=1, F=0?

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0			

Relaxed Consistency

Possible with SC: A=0, B=0, C=1, D=0, E=1, F=0? **No**

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
A = *y;
```

```
// P2
```

```
*y = 1;
B = *x;
```

```
// P3
```

```
C = *x;
D = *y;
```

```
// P4
```

```
E = *y;
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no		

Relaxed Consistency

Possible with TSO: A=0, B=0, C=1, D=0, E=1, F=0?

- Relaxed write-to-read, with store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no		

Relaxed Consistency

Possible with TSO: A=0, B=0, C=1, D=0, E=1, F=0?

- Relaxed write-to-read, with store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

relaxed write to read

WAR

```
// P1          // P2
*x = 1;        *y = 1;
A = *y;        B = *x;
```

```
// P3
C = *x;
D = *y;
```

```
// P4
E = *y;
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no		

Relaxed Consistency

Possible with TSO: A=0, B=0, C=1, D=0, E=1, F=0?

- Relaxed write-to-read, with store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
RAW C = *x;
```

```
WAR D = *y;
```

```
// P4
```

```
RAW E = *y;
```

```
WAR F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no		

Relaxed Consistency

Possible with TSO: A=0, B=0, C=1, D=0, E=1, F=0? **No**

➤ Relaxed write-to-read, with store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no	no	

Relaxed Consistency

Possible with PC: A=0, B=0, C=1, D=0, E=1, F=0?

- Relaxed write-to-read, without store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no	no	

Relaxed Consistency

Possible with PC: A=0, B=0, C=1, D=0, E=1, F=0?

- Relaxed write-to-read, without store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

relaxed write to read

```
// P1
```

```
*x = 1;
A = *y;
```

```
// P2
```

```
*y = 1;
B = *x;
```

WAR

without store atomicity:
can disagree on order of
writes to x and y

```
// P3
```

```
C = *x;
D = *y;
```

```
// P4
```

```
E = *y;
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no	no	



Relaxed Consistency

Possible with PC: A=0, B=0, C=1, D=0, E=1, F=0?

- Relaxed write-to-read, without store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

without store atomicity:

*x = 1;



*y = 1;



*x = 1;

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no	no	



Relaxed Consistency

Possible with PC: A=0, B=0, C=1, D=0, E=1, F=0?

- Relaxed write-to-read, without store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

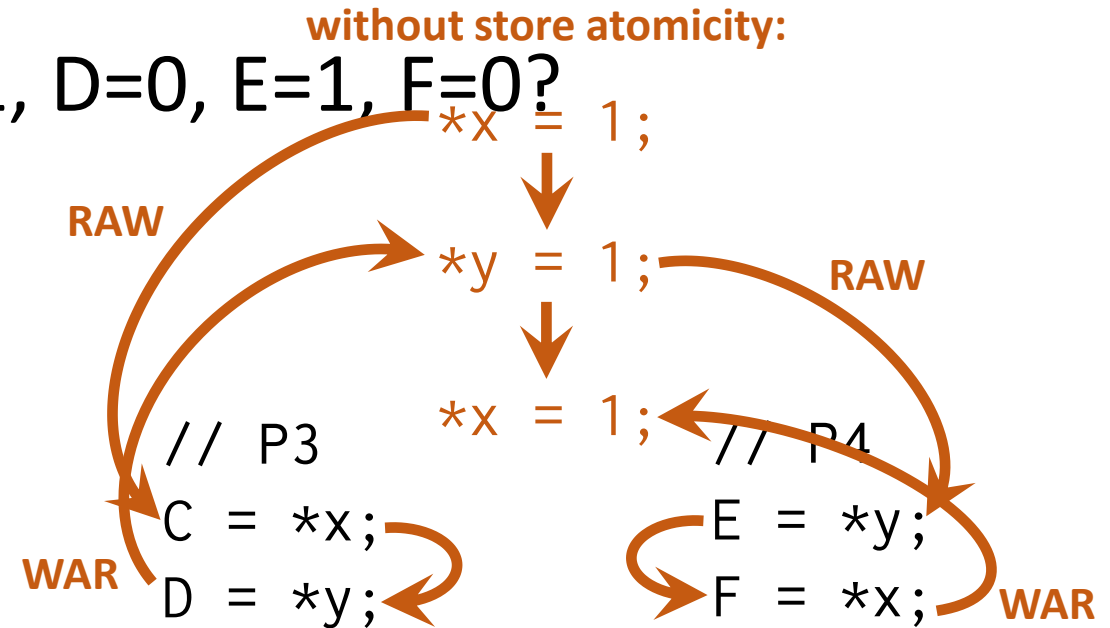
```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```



	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no	no	

Relaxed Consistency

Possible with PC: A=0, B=0, C=1, D=0, E=1, F=0? **Yes**

➤ Relaxed write-to-read, without store atomicity

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no	no	yes



Relaxed Consistency

Possible with SC: A=0, B=1, C=0, D=1, E=0, F=1?

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no	no	yes
iii.	0	1	0	1	0	1			



Relaxed Consistency

Possible with SC: A=0, B=1, C=0, D=1, E=0, F=1? **Yes**

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

happens last

```
// P3
```

happens first

```
// P4
```

```
C = *x;
```

```
D = *y;
```

```
E = *y;
```

```
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no	no	yes
iii.	0	1	0	1	0	1	yes		



Relaxed Consistency

Possible with TSO/PC: A=0, B=1, C=0, D=1, E=0, F=1? Yes

```
// init
```

```
*x = 0; *y = 0;
```

```
// P1
```

```
*x = 1;
```

```
A = *y;
```

```
// P2
```

```
*y = 1;
```

```
B = *x;
```

```
// P3
```

```
C = *x;
```

```
D = *y;
```

```
// P4
```

```
E = *y;
```

```
F = *x;
```

	A	B	C	D	E	F	SC?	TSO?	PC?
i.	0	0	0	0	0	0	no	yes	yes
ii.	0	0	1	0	1	0	no	no	yes
iii.	0	1	0	1	0	1	yes	yes	yes

